SatarkPay · SPHINX'26, MNIT Jaipur · PS 3.2

# Testing Guide

How to test every part: website, phone, QR, payments, SMS, email, browser extension, security and the demo.

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Everything in SatarkPay is simulated. How it would do on real payments is unknown.

# How to use this guide

This is the checklist for testing every part of SatarkPay before the submission and the demo. Work through it section by section and write Pass or Fail in the last column. Scores can move by a few points between runs, so the risk level matters more than the exact number.

| In the table | Meaning |
|---|---|
| **Tested** | Passed when run by the automated tests or by hand on 30 and 31 Oct 2026. Still repeat it on the demo laptop and phone. |
| **Simulated only** | Passed, but only against a stand-in, such as a simulated Gmail page. |
| **Not yet: needs you** | Nobody has confirmed this. It needs a real person, a real phone or a real account. **Do these first.** |

### The highest-risk tests

> **Do these five first:** Q1 (live camera), Q7 (payment app with ₹1), S2 (real scam SMS), M2 (real Gmail) and D1 (full rehearsal). None has been confirmed yet, and they are the ones most likely to surprise you on stage.

# 1. Set up

### Laptop

```
git clone https://github.com/ajayyadav432/SatarkPay.git
cd SatarkPay
./run.sh            # first run installs everything and prints the login passwords once
# open http://localhost:8000
```

### Phone (Android, USB debugging on)

```
cd android && ./gradlew assembleDebug
adb install -r --user 0 app/build/outputs/apk/debug/app-debug.apk   # --user 0 was needed on our Samsung
./run.sh phone      # in a second terminal: keeps the phone connected over USB
```

In the app, the address under **Laptop connection** is `http://127.0.0.1:8000` over USB. Over Wi-Fi, start with `./run.sh wifi` and use the address and pairing code it prints.

### Browser extension

- Open `chrome://extensions` and turn on **Developer mode**.
- Press **Load unpacked** and choose the `extension` folder.
- Pin **SatarkPay Link Guard**. It only talks to `127.0.0.1:8000`, so keep `./run.sh` running.

### Test logins

Every portal has a **Use test login** button, on the website and in the app. No password is needed. Use these for all tests below unless a test says otherwise.

# 2. Automated tests

Run these first. They should all pass, except 1 skipped test (it covers a page that is not part of SatarkPay).

| What | Command | Tests |
|---|---|---|
| Everything on the laptop | `./run.sh test` | 561 |
| Scoring, features, simulator | `./run.sh test tests/test_scorer.py tests/test_features.py tests/test_simulate.py` | 165 |
| API and security | `./run.sh test tests/test_api.py` | 71 |
| Login, sign-up, roles | `./run.sh test tests/test_auth.py` | 71 |
| Problem statement requirements | `./run.sh test tests/test_acceptance.py` | 12 |
| Website in a browser, and extension | `./run.sh test tests/test_e2e.py` | 44 |
| Text and link engine | `./run.sh test tests/engine` | 183 |
| Documents match the measured numbers | `./run.sh test tests/test_docs.py` | 13 |
| Phone: QR parser (no phone needed) | `cd android && ./gradlew testDebugUnitTest` | 13 |
| Phone: real device (server running) | `adb shell am instrument -w --user 0 ai.satark.smsguard.test/androidx.test.runner.AndroidJUnitRunner` First: `./gradlew assembleDebugAndroidTest`, then `adb install -r -t --user 0` with the APK in `app/build/outputs/apk/androidTest/debug/` | 15 |

> The browser tests need Playwright: `.venv/bin/python -m playwright install chromium`. The last run: 560 of 561 passed on the laptop with 1 skipped, 44 of 44 in the browser, and 15 of 15 on the Samsung SM-E366B.

# 3. Login and portals

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| L1 | Open the website without logging in, at `/` | You are sent to the login page. | **Tested** | |
| L2 | Press **Use test login** on User, then Merchant, then Admin. | User opens `/pay`. Merchant opens `/merchant`. Admin opens the dashboard. | **Tested** | |
| L3 | Sign up as a new user, with a password of 8 or more characters. | You land on the user screen, and your name is on the account. | **Tested** | |
| L4 | Sign up as a merchant with a shop name. | You land on the merchant screen with your shop name. | **Tested** | |
| L5 | Sign up with a username already taken, and with a 3-character password. | "That username is taken." and a message about password length. | **Tested** | |
| L6 | Choose Admin on the login page. | No sign-up option. "Admin accounts cannot be created here." | **Tested** | |
| L7 | Enter a wrong password. | "Wrong username or password." It does not say which part was wrong. | **Tested** | |
| L8 | Enter a wrong password 6 times on one account. | "Too many wrong tries." The account locks for a minute. Others still work. | **Tested** | |
| L9 | Log in as User, then open `/` and `/merchant`. | You are returned to `/pay`. | **Tested** | |
| L10 | Log out. | Back to login. The old session no longer works. | **Tested** | |
| L11 | Repeat L2, L3 and L7 in the phone app. | The same behaviour. The app has its own home screen for each role. | **Tested** (L3 and L7 partly) | |

# 4. Website: user and merchant

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| U1 | User: tap **Pay the local shop**. | "Looks fine". Pay goes through and shows Paid. | **Tested** | |
| U2 | User: tap **A "refund" request**. | Stopped, with reasons, and 1930 shown. | **Tested** | |
| U3 | On U2, switch to Hindi. | Title and reasons in Hindi. | **Tested** | |
| U4 | On U2, press Report. | "This account now has N reports." | **Tested** | |
| U5 | Run all 7 ready-made scenarios. | Shop is allowed. The other 6 warn, hold or block, with reasons. | **Tested** | |
| U6 | **Beat the model**: enter a UPI ID, amount and a note. If warned, try Pay anyway. | A decision appears. Pay anyway is locked until the pause ends. | **Tested** | |
| U7 | Beat the model: put `<img src=x onerror=alert(1)>` in the note. | It shows as plain text. No popup. | **Tested** | |
| U8 | Beat the model: enter a bad UPI ID such as `not an id`. | A clear message, not a crash. | **Tested** | |
| MR1 | Merchant: press **A customer pays**. | A new row showing Received, and the total goes up. | **Tested** | |
| MR2 | Check the merchant screen again after the live stream has run for a while. | The total stays. It does not fall back to ₹0. | **Tested** | |

# 5. Website: admin dashboard

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| A1 | Open the dashboard. | "Live", with the feed moving and the numbers rising. | **Tested** | |
| A2 | Launch each of the 5 attacks. | Each card reports payments and how many were flagged and stopped. | **Tested** | |
| A3 | Click an alert. | Reasons, SHAP bars and a money network. | **Tested** | |
| A4 | Press **Confirm scam** on an alert. | It changes to Confirmed scam. | **Tested** | |
| A5 | Pause, then change the speed. | "Paused". The feed stops, then runs at the new speed. | **Tested** | |
| A6 | Open a phone at `/pay` and launch Fake collect. | The request appears on the open phone. | **Tested** | |
| A7 | Open `/how`. | Measured numbers and limits. No NaN or undefined. | **Tested** | |
| A8 | Open every page in a phone-sized window. | Nothing scrolls sideways. | **Tested** | |

# 6. Phone app: QR and payment

Open `http://localhost:8000/qr` on the laptop. It has 5 test codes.

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| Q1 | **Press Scan UPI QR and scan "Scan to receive" with the live camera.** | The phone vibrates and shows the payment card with ₹12,000 and "scan to receive". | **Not yet: needs you** | |
| Q2 | Press Check before paying on Q1. | Orange **On hold**, about risk 78. Pay anyway locked for 15 s. | **Tested** (from a picture) | |
| Q3 | Scan "A local shop". | Green "Looks fine", risk about 0. Simulate payment works. | **Tested** (from a picture) | |
| Q4 | Scan "A fake refund" and "A job deposit". | Fake refund: Block, about 85. Job deposit: Hold, about 66. | **Not yet: needs you** | |
| Q5 | Scan "A QR with a fake bank link". | "This QR is a web link". Red: Do not open or pay. About Critical 88. | **Tested** (from a picture) | |
| Q6 | Scan a real shop QR, or your own UPI QR. | Grey "No warning signs found", and a note that it is not a guarantee. Never green or Safe. | **Tested** (from a picture) | |
| Q7 | **On your own UPI ID with ₹1, press Open payment app. Check the details, then pay.** | Your UPI app opens with the right ID and amount. You enter your PIN there. **Real ₹1 moves.** Some apps may refuse the request. | **Not yet: needs you** | |
| Q8 | Use a blocked payment (Q4 fake refund). | No payment button. Report and 1930 are offered. | **Not yet: needs you** | |
| Q9 | **From photo**: choose a screenshot of a QR. | The same payment card as Q1. | **Not yet: needs you** | |
| Q10 | From photo: choose a picture with no QR. | "No QR code found in that picture." | **Not yet: needs you** | |
| Q11 | In WhatsApp or Gallery, share a QR picture to SatarkPay. | The app opens the payment card. If it is not logged in, log in first. | **Not yet: needs you** | |
| Q12 | Scan any QR that is not a payment or link (a Wi-Fi code, say). | "That QR code is not a UPI payment or a web link". | **Tested** (parser) | |
| Q13 | **Enter UPI ID**: type an ID, amount and note, and check. | A decision appears. Bad input gets a clear message. | **Not yet: needs you** | |
| Q14 | Turn the server off, then press Check. | "The laptop did not answer. Do not pay until you can check." | **Not yet: needs you** | |

# 7. SMS

> **Privacy:** once SMS is allowed, every SMS the phone receives, including OTPs and bank messages, goes to the laptop while the server runs. Use a test phone if you can, and remove the permission afterwards. Messages are never sent to an `https` address such as the Vercel copy. Do not open the links in these samples.

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| S1 | Laptop connection: press **Try a sample scam SMS**. | Critical 98, plus a notification. | **Tested** | |
| S2 | **Allow SMS, then send sample 1 below from a second phone.** | A notification, and Critical in Recently checked. | **Not yet: needs you** | |
| S3 | Send samples 2 and 3 from the second phone. | Safe. No alarm. | **Not yet: needs you** | |
| S4 | Send samples 4, 5 and 6. | Critical, with a category named for each. | **Not yet: needs you** | |
| S5 | Send 3 scam messages from one number. | A recommendation to block that sender. | **Not yet: needs you** | |
| S6 | Turn the server off and send a message. | "Not checked". After restarting, a Retry button appears under Laptop connection. | **Not yet: needs you** | |
| S7 | Set the app address to the Vercel link and send a message. | Nothing is sent and no check happens. Set the address back afterwards. | **Not yet: needs you** | |
| S8 | On the home screen, press **Check message** and paste sample 1. | Red "Critical risk. Do not reply, click or pay." | **Tested** | |

### Samples and the results measured on 31 Oct 2026

| # | Message | Result |
|---|---|---|
| 1 | Dear customer, your SBI account will be blocked today. Update your KYC immediately: http://sbi-kyc-update.xyz/login | Critical 98, KYC |
| 2 | 482913 is your OTP for transaction of INR 2,499.00 at AMAZON. Do not share it with anyone. -HDFC Bank | Safe 0 |
| 3 | Hey are we still meeting for lunch tomorrow at 1? Let me know | Safe 0 |
| 4 | Congratulations! You have won Rs 25,00,000 in KBC lucky draw. To claim your prize pay processing fee Rs 5,000 to kbcoffice@ybl. Call 8765432109 | Critical 89, prize |
| 5 | Priya grahak, aapka HDFC khata band ho jayega. Turant KYC update karein, link par click karein: http://hdfc-kyc-help.top/login | Critical 98, KYC |
| 6 | This is Inspector Sharma from CBI. Your Aadhaar is linked to a money laundering case. You are under digital arrest. Join the video call now and do not tell anyone. Transfer Rs 2,00,000 for verification. | Critical 88, digital arrest |

# 8. Email (Gmail)

The extension reads an opened email in **Gmail in a desktop browser** and shows a bar above it. It does not cover the Gmail phone app, Outlook or other mail sites. The email text goes only to the SatarkPay server on the same laptop. Gmail does not document its page structure, so if Gmail changes it, the bar simply will not appear.

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| M1 | Automated: a simulated Gmail page with sample A. | A bar reading Critical 99. | **Simulated only** | |
| M2 | **From a second account, send yourself sample A. Open it in Gmail in Chrome, with the extension on and the server running.** Check Spam if it does not arrive. | A bar above the email: Critical, with reasons such as "Sender address imitates..." if the sender looks that way. | **Not yet: needs you** | |
| M3 | Send and open sample B. | "no strong scam signs found." | **Not yet: needs you** | |
| M4 | Turn off protection in the extension, and reload the email. | No bar. | **Not yet: needs you** | |
| M5 | Stop the server, and open an email. | No bar and no error. Gmail keeps working. | **Not yet: needs you** | |
| M6 | Send several scam emails from one sender. | The bar recommends blocking that sender. | **Not yet: needs you** | |

| | Email text | Measured |
|---|---|---|
| A | Your SBI net banking will be suspended in 24 hours. Verify your account now at http://sbi-secure-alerts.xyz/verify | Critical 99 with the fake sender `support@sbi-secure-alerts.xyz` |
| B | Sharing the notes from yesterday's class. See you on Monday. | Safe 0 |

> Your real test mail will come from a normal address, so its score can be lower than 99, because the sender no longer looks fake. Note: the bar has no Details link in SatarkPay, because there is no case page. That is expected.

# 9. Browser extension

Start the test pages in a second terminal: `./run.sh demo`, then open `http://127.0.0.1:8765`. All names and offers on them are made up.

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| E1 | Click the extension icon. | Protection on, and connected to SatarkPay on this computer. | **Tested** (Chromium) | |
| E2 | Open **Demo inbox**. | 2 red Critical badges on the scam links. None on the Wikipedia link. | **Tested** (Chromium) | |
| E3 | Click the SBI scam link. | A warning page, with Go back and Open the link anyway. | **Tested** (Chromium) | |
| E4 | On the warning page, press Go back. | You return to the inbox. | **Not yet: needs you** | |
| E5 | Open **Chatter**, the fake social network. | Several badges on scam posts (10 when tested). | **Tested** (Chromium) | |
| E6 | Click a real link such as Wikipedia. | It opens with no warning. | **Not yet: needs you** | |

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| E7 | Turn the switch off in the popup, and reload. | Badges disappear, and clicks are not checked. | **Not yet: needs you** | |
| E8 | Stop the server, and browse. | Nothing is blocked. The popup says the app is not running. | **Not yet: needs you** | |
| E9 | Load the extension in your own Google Chrome. | The same as above. Automated runs used Chromium, so this has not been checked in Chrome itself. | **Not yet: needs you** | |

# 10. Link checks

Use **Check link** in the app, or a browser at `/api/check-url?u=LINK`. Measured on 31 Oct 2026.

| Link | Result | Note |
|---|---|---|
| `http://sbi-kyc-update.xyz/login` | Critical 90 | Fake bank page |
| `https://www.onlinesbi.sbi` | Safe 0 | The real site |
| `https://www.google.com` | Safe 0 | |
| `https://evil.com\@sbi.co.in/` | Medium 55 | Looks like SBI but opens evil.com |
| `paytm.me/abc` | High 72 | **A known false alarm** in the engine on its own. Inside a payment, the context layer corrects for it |

# 11. Security

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| X1 | Without logging in, run `curl localhost:8000/api/v1/kpis`. | 401. | **Tested** | |
| X2 | Logged in as User, open `/api/v1/kpis`. | 403. | **Tested** | |
| X3 | Logged in as User, try to listen to the shop's stream. | 403. | **Tested** | |
| X4 | Change one character of a session token. | 401. | **Tested** | |
| X5 | Send a request with a different Host header, or from another website's Origin. | 403. | **Tested** | |
| X6 | Look at `.env` and the Git history for passwords. | Only hashes in `.env`, and `.env` is not in Git. | **Tested** | |
| X7 | Try `https://satarkpay.vercel.app/.env`. | 404. | **Tested** | |

# 12. Vercel copy

At `https://satarkpay.vercel.app`. This is the backup for the laptop, not the main demo. Sign-ups, alerts and live numbers can reset or differ between visits.

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| V1 | Use each of the 3 test logins. | Each portal opens. | **Tested** | |
| V2 | User: run the fake refund. | Block. | **Tested** | |
| V3 | Merchant: press A customer pays. | A row appears. | **Tested** | |
| V4 | Admin: launch Mule ring. | An alert appears within about 45 seconds. | **Tested** | |
| V5 | Sign up, then come back later. | The account may be gone. That is expected. | **Tested** (sign-up) | |
| V6 | Open `/how` and `/qr`. | The page and all 5 QR codes load. | **Tested** | |
| V7 | **Try it on the venue Wi-Fi and on mobile data.** | It works on both. | **Not yet: needs you** | |
| V8 | In the app, set the address to the Vercel link and scan a test QR. | The check works over the internet. | **Not yet: needs you** | |

# 12b. Scan, feed, history, languages and connection

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| N1 | On /pay choose From a photo and pick a test QR. | The steps run, with a progress bar and a Skip button, then the decision. | **Tested** | |
| N2 | On /pay press Scan a QR on the online copy with a phone browser. | The camera opens and reads a QR. Needs https or localhost. | **Not yet: needs you** | |
| N3 | Post on /feed naming an ID and a phone number. | The phone number is hidden. A real ID is partly hidden. | **Tested** | |
| N4 | Report one simulated ID from 3 accounts, then pay it. | A warning that says the reports are not verified. | **Tested** | |
| N5 | As admin, block it from the most reported list, then pay it. | Blocked. | **Tested** | |
| N6 | On /history filter, then download CSV and JSON. | Both files download. Open the CSV in Excel: rupee signs and Hindi read correctly. | **Tested** | |
| N7 | Switch through all 6 languages on a decision. | Buttons, decision and reasons change. A native speaker checks the wording. | **Not yet: needs you** | |
| N8 | In the app open Connection and choose Online, Wi-Fi and USB in turn. | Test connection works for each. | **Not yet: needs you** | |
| N9 | In the app, on a real UPI ID with a small amount, tap one of the listed UPI apps. | That app opens with the details filled in. Do not enter a PIN unless you mean to pay. | **Not yet: needs you** | |

# 13. Demo rehearsal and failure drills

| ID | Do this | You should see | Checked so far | Pass / fail |
|---|---|---|---|---|
| D1 | **Follow docs/DEMO_SCRIPT.md with a timer, 3 times.** | Finishes on time. Everyone knows their part. | **Not yet: needs you** | |
| D2 | Pull out the USB cable mid-demo. | The app says the laptop is offline. Reconnect with `adb reverse tcp:8000 tcp:8000`. | **Not yet: needs you** | |
| D3 | Make the camera unusable. | Scan from a saved QR screenshot using From photo. | **Not yet: needs you** | |
| D4 | Close the laptop server. | Switch to the Vercel link and say that it is the test copy. | **Not yet: needs you** | |
| D5 | Remove internet. | The laptop demo still works. Only Vercel stops. | **Not yet: needs you** | |
| D6 | Have someone ask questions from docs/JUDGING_QA.md. | Everyone can answer in their own area. | **Not yet: needs you** | |
| D7 | Open the deck in PowerPoint, on the presenting laptop. | All 12 slides look right, including the fonts. | **Not yet: needs you** | |

# 14. Before the demo

- Run `./run.sh test` once more on the demo laptop.
- Press **Clear** under Recently checked in the app.
- Restart the server, so the numbers start from zero.
- Charge the phone, and set its screen to stay on.
- Save a QR screenshot on the phone as a backup.
- Remove the SMS permission if you are not demonstrating SMS.

## If a test fails

Write down the test ID, what you did, what you saw and a screenshot. Then open a GitHub issue or tell Ajay. Do not push a quick fix straight to `main` just before the demo. Use a pull request, and run the tests first.

## Sign-off

| Section | Tested by | Date | Result |
|---|---|---|---|
| Automated tests | | | |
| Login and portals | | | |
| Website | | | |
| Phone: QR and payment | | | |
| SMS | | | |
| Email | | | |
| Extension | | | |
| Vercel | | | |
| Rehearsal | | | |